Multivariate multicycle codes: complete code tables

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Complete versions of Tables II–XI of "Multivariate Multicycle Codes for Complete Single-Shot Decoding". The paper lists a selection of these instances. An entry $\leq d$ denotes an upper bound given by the weight of a verified nontrivial logical operator; an entry without $\leq$ denotes an exact distance established by matching upper and lower bounds (Appendix C of the paper and the `Distance Audit` directory of this repository).

$[[n,k,(\leq d_X,\leq d_Z)]]$	(ℓ,m,p,q)	k/n	kd^2/n	F	G	H	I
$[[96,12,(4,4)]]$	(2, 2, 2, 2)	0.125	2.00	$wz+xyz$	$w+xyz$	$wyz+xy$	$xyz+xz$
$[[144,6,(8,8)]]$	(2, 2, 3, 2)	0.042	2.67	$wxy+z$	$wx+wyz$	$wx+xy^2$	$xyz+y^2z$
$[[216,6,(\leq 12,\leq 12)]]$	(2, 2, 3, 3)	0.028	4.00	$wxy+xy^2$	$wy+xyz^2$	xy^2+y^2z	$wxyz^2+xz$
$[[240,12,(\leq 7,\leq 7)]]$	(5, 2, 2, 2)	0.050	2.45	w^4y+w^3x	w^4x+w^3	w^4y+wx	w^4z+w^2yz
$[[240,6,(\leq 12,\leq 12)]]$	(5, 2, 2, 2)	0.025	3.60	w^4xy+xz	w^3+w^2z	$w^3xyz+wy$	w^2y+w
$[[288,6,(\leq 12,\leq 12)]]$	(2, 2, 3, 4)	0.021	3.00	$wxy+xz^2$	wxz^2+wy^2	xyz^3+z^2	$wxy+w$
$[[288,12,(8,8)]]$	(2, 2, 2, 6)	0.042	2.67	wz^3+xyz^2	wxz^2+xyz	wxz^2+w	wyz^5+xyz^3
$[[324,6,(\leq 12,\leq 12)]]$	(2, 3, 3, 3)	0.019	2.67	$wxz+yz^2$	$wxyz+xz$	$wx^2z+y^2z^2$	wxy^2+wxz^2
$[[336,6,(\leq 12,\leq 12)]]$	(2, 2, 2, 7)	0.018	2.57	$wx+z^2$	wxz^2+z^6	$wx+wz^2$	wxz^4+wyz^6
$[[360,6,(\leq 12,\leq 12)]]$	(2, 2, 3, 5)	0.017	2.40	xz^3+y^2z	$wx+wz^2$	$wxy^2+y^2z^2$	$wxyz^3+wxz^2$
$[[360,12,(\leq 11,\leq 11)]]$	(2, 2, 3, 5)	0.033	4.03	y^2z+yz^3	$wy^2z^3+xy^2$	wz^3+xyz^2	wyz^3+xy^2
$[[432,12,(12,12)]]$	(2, 2, 2, 9)	0.028	4.00	wyz^3+xz^7	$wyz+yz^5$	xyz^5+z^7	wxz^7+wyz^3
$[[432,6,(\leq 18,\leq 18)]]$	(2, 3, 3, 4)	0.014	4.50	$wx+x^2z$	wy^2z^3+xyz	x^2yz+y^2	xy^2z^3+xyz
$[[480,6,(\leq 16,\leq 16)]]$	(2, 2, 2, 10)	0.013	3.20	$w+xz^8$	wxz^7+w	wxz^7+xyz^3	$wyz+wz^7$
$[[480,12,(\leq 12,\leq 12)]]$	(2, 2, 4, 5)	0.025	3.60	$wxy^2z^4+wz^2$	$wxz+xy^2z^4$	$wxy^2z^2+xy^2$	xy^3z+xyz^3
$[[480,24,(\leq 8,\leq 8)]]$	(2, 2, 4, 5)	0.050	3.20	wxy^3z^2+xz	wz^3+yz	$wy^2z^3+y^3z^4$	wyz^4+y^2z
$[[504,6,(\leq 17,\leq 17)]]$	(2, 2, 3, 7)	0.012	3.44	$wxyz^4+wz$	$wy^2z^3+xz^3$	$wy^2z^5+xy^2z$	wyz^2+yz
$[[504,12,(\leq 10,\leq 10)]]$	(2, 2, 3, 7)	0.024	2.38	$wx+xyz^4$	wyz^2+y	wy^2z^4+wyz	wyz^5+z^2
$[[528,6,(\leq 16,\leq 16)]]$	(2, 2, 2, 11)	0.011	2.91	xyz^5+z^2	$y+z^9$	wxz^3+xyz^7	$wyz^{10}+xyz^4$
$[[528,12,(\leq 12,\leq 12)]]$	(2, 2, 2, 11)	0.023	3.27	$wz^{10}+wz^9$	$yz^{10}+z^6$	wz^5+xyz	$wxyz^6+yz^{10}$
$[[540,6,(\leq 18,\leq 18)]]$	(2, 3, 3, 5)	0.011	3.60	$wx^2yz+wxy^2z^2$	$xz+1$	$wz+x^2yz^2$	wx^2yz+wx
$[[540,18,(\leq 7,\leq 7)]]$	(2, 3, 3, 5)	0.033	1.63	$wxy^2z^2+x^2y^2z^2$	$wxyz^2+xy$	x^2y+xyz^2	$x^2y^2+y^2z^2$
$[[600,6,(\leq 17,\leq 17)]]$	(2, 2, 5, 5)	0.010	2.89	xy^2+xz	wz^4+yz^4	$xz^2+y^2z^3$	wxz^4+y^2z
$[[600,24,(\leq 9,\leq 9)]]$	(2, 2, 5, 5)	0.040	3.24	y^4z^3+yz	$xy^3+xy^2z^4$	$wy^3z^4+wy^3$	y^3z^2+yz
$[[600,12,(\leq 10,\leq 10)]]$	(2, 2, 5, 5)	0.020	2.00	y^3z^3+yz	$wyz+xy^3z$	$wy^3z^2+wy^2z^2$	wy^2z+wy^2
$[[624,6,(\leq 18,\leq 18)]]$	(2, 2, 2, 13)	0.010	3.12	xz^6+yz^9	xyz^7+xz	$wxyz^{11}+wyz^{12}$	$wyz^{10}+xz^5$
$[[624,12,(\leq 14,\leq 14)]]$	(2, 2, 2, 13)	0.019	3.77	$wyz^{12}+xz^7$	wz^9+wz	xz^11+yz^8	$wxyz^3+xyz^4$
$[[648,6,(\leq 18,\leq 18)]]$	(2, 3, 3, 6)	0.009	3.00	wx^2y^2z+xz	$wyz^2+y^2z^2$	$w+x^2yz$	$wxyz^4+xy^2$
$[[648,12,(12,12)]]$	(2, 2, 3, 9)	0.019	2.67	$xy^2z^4+xy^2z^3$	$wx+xz^8$	$wz^7+y^2z^2$	$wy^2z^2+z^5$
$[[672,6,(\leq 18,\leq 18)]]$	(2, 2, 4, 7)	0.009	2.89	wxy^3z^3+w	$wy^3z+y^2z^4$	xyz^2+xz^3	$xy+y^3z^2$
$[[672,12,(\leq 12,\leq 12)]]$	(2, 2, 2, 14)	0.018	2.57	$wz^{13}+xz^9$	$wxyz^5+yz^{13}$	wxz^8+wyz^{11}	wxz^8+yz^{11}
$[[672,24,(\leq 11,\leq 11)]]$	(2, 2, 4, 7)	0.036	4.32	wy^3z+wy^3	$wxyz^4+wz^6$	y^2z^2+1	wxy^3+wz^2
$[[720,6,(\leq 20,\leq 20)]]$	(2, 2, 5, 6)	0.008	3.33	$wxy^2z^4+xyz^4$	wxy^4z+yz^3	$wxyz^5+xyz^4$	wxy^2z^3+yz
$[[720,12,(\leq 13,\leq 13)]]$	(2, 2, 2, 15)	0.017	2.82	wxz^6+wy	xyz^3+y	xyz^7+xz^5	yz^5+z^4
$[[720,12,(\leq 15,\leq 15)]]$	(2, 2, 3, 10)	0.017	3.75	wxz^8+wxz^6	wyz^5+wz^2	$wxz^6+xy^2z^6$	wxz^5+xyz^6
$[[756,6,(\leq 18,\leq 18)]]$	(2, 3, 3, 7)	0.008	2.57	wy^2z+wyz^6	wxz^3+x^2	$x^2yz^5+xz^6$	yz^3+z^5
$[[756,18,(\leq 13,\leq 13)]]$	(2, 3, 3, 7)	0.024	4.02	$wx^2y^2z^2+x^2y^2$	$wy^2+x^2yz^5$	$wx^2z^2+x^2z^3$	$x^2yz^3+xz^4$
$[[792,6,(\leq 22,\leq 22)]]$	(2, 2, 3, 11)	0.008	3.67	$wxyz^2+wxz$	$xyz^5+y^2z^7$	wxz^4+z^3	$xy^2z^3+y^2z$
$[[792,12,(\leq 14,\leq 14)]]$	(2, 2, 3, 11)	0.015	2.97	$wxyz^3+wxyz$	$wx+y^2z^3$	$wxyz^9+y^2z^{10}$	$wz^5+xy^2z^5$
$[[792,12,(\leq 15,\leq 15)]]$	(2, 2, 3, 11)	0.015	3.41	$wz+z^{10}$	$wxy^2z^4+xyz^7$	$xy^2z^9+xz^{10}$	wyz^6+wyz^5
$[[810,18,(\leq 12,\leq 12)]]$	(3, 3, 3, 5)	0.022	3.20	$w^2x^2y^2+w^2x^2yz^3$	$w^2xyz+yz$	w^2yz^4+wxz	$x^2y^2z^3+x^2y^2z$
$[[810,6,(\leq 21,\leq 21)]]$	(3, 3, 3, 5)	0.007	3.27	$w^2xy^2z^3+y^2z$	$wz+xy^2z^2$	$wyz^4+x^2yz^2$	wx^2y+wxy^2z

TABLE II (continued)

$[[n, k, (\leq d_X, \leq d_Z)]]$	(ℓ, m, p, q)	k/n	kd^2/n	F	G	H	I
$[[816, 6, (\leq 20, \leq 20)]]$	(2, 2, 2, 17)	0.007	2.94	$wxyz^4 + z^9$	$xyz^{16} + xyz^2$	$wxyz + x$	$wxyz^9 + wxz^5$
$[[816, 12, (\leq 12, \leq 12)]]$	(2, 2, 2, 17)	0.015	2.12	$yz^{15} + yz$	$wyz^7 + z^6$	$wyz^6 + 1$	$wxz^2 + xz^3$
$[[816, 24, (\leq 10, \leq 10)]]$	(2, 2, 2, 17)	0.029	2.94	$wxz^2 + 1$	$wxyz^4 + yz^8$	$wz^8 + xz^7$	$wxyz^4 + yz$
$[[840, 6, (\leq 20, \leq 20)]]$	(2, 2, 5, 7)	0.007	2.86	$wxy^4z + wy^4z^5$	$wy + z^5$	$wxz^2 + xy^4z^2$	$wxy^4 + wy^3z^3$
$[[840, 24, (\leq 10, \leq 10)]]$	(2, 2, 5, 7)	0.029	2.86	$y^4 + y^3z^2$	$wxy^3z^4 + wxy^2z^2$	$wy^3z^6 + wy^3$	$y^4z^3 + y^3z^2$
$[[840, 12, (\leq 14, \leq 14)]]$	(2, 2, 5, 7)	0.014	2.80	$wxy^3z + wyz^6$	$wxyz^4 + wxz^3$	$xz^4 + xz^3$	$xy^3z + y^2z^4$
$[[864, 6, (\leq 24, \leq 24)]]$	(2, 2, 6, 6)	0.007	4.00	$wy^2z^4 + wyz^3$	$xy^5z^3 + xy^4z^3$	$xy^3z + y^5z^5$	$wz^5 + 1$
$[[864, 12, (\leq 14, \leq 14)]]$	(2, 2, 2, 18)	0.014	2.72	$wz^3 + xyz^7$	$wxyz^{14} + xyz^6$	$wxyz^7 + yz^{14}$	$wyz + xz^{17}$
$[[864, 18, (\leq 12, \leq 12)]]$	(2, 3, 4, 6)	0.021	3.00	$wx^2y^3z^4 + wz^2$	$wx^2yz^4 + xy^2$	$x^2y^3 + y^2z$	$wx^2y^3z + wxy^3z^3$
$[[864, 24, (\leq 10, \leq 10)]]$	(2, 2, 2, 18)	0.028	2.78	$wxyz^6 + yz^8$	$wxyz^{12} + yz^5$	$z^4 + z^2$	$wxz^9 + z^6$
$[[900, 6, (\leq 24, \leq 24)]]$	(2, 3, 5, 5)	0.007	3.84	$x^2y^4z^4 + x^2y^2z$	$wx^2y^4z^3 + wy^4z^2$	$wx^2z^2 + z^4$	$wxy^3 + wxz^2$
$[[900, 12, (\leq 11, \leq 11)]]$	(2, 3, 5, 5)	0.013	1.61	$wxz^4 + wyz^3$	$x^2y^4z^3 + x^2yz^2$	$wy^3z^3 + wy^3z$	$x^2y^3z^4 + xy$
$[[912, 12, (\leq 16, \leq 16)]]$	(2, 2, 2, 19)	0.013	3.37	$w + z^{10}$	$w + z^{11}$	$wyz^{12} + yz^{10}$	$wyz^{12} + w$
$[[936, 6, (\leq 25, \leq 25)]]$	(2, 2, 3, 13)	0.006	4.01	$wz^{12} + y^2z^{11}$	$wz^3 + z^5$	$wy^2z^{11} + z^2$	$wx + wy^2z^2$
$[[936, 12, (\leq 16, \leq 16)]]$	(2, 2, 3, 13)	0.013	3.28	$wy^2z^2 + y^2z^{11}$	$y^2z^6 + z^{11}$	$wy^2z^2 + wyz^6$	$xy^2z^6 + xz^3$
$[[972, 36, (\leq 9, \leq 9)]]$	(2, 3, 3, 9)	0.037	3.00	$wx^2z^8 + wxy^2z^5$	$wxz^6 + wyz^3$	$x^2y^2z^5 + z^5$	$wx^2y^2 + wy^2$
$[[972, 18, (\leq 13, \leq 13)]]$	(2, 3, 3, 9)	0.019	3.13	$wx^2y^2z^6 + wx^2yz$	$wx^2y^2z^7 + x^2yz^3$	$wxy^2z^8 + xyz$	$wxy^2z^5 + wxy^2z^3$
$[[972, 6, (\leq 24, \leq 24)]]$	(2, 3, 3, 9)	0.006	3.56	$x^2z^5 + xy^2z^3$	$wz^3 + y^2z^8$	$wx^2z^4 + wxz^2$	$xyz^4 + y^2z^6$

TABLE II: **Small instances of multivariate multicycle (MM) codes with stabilizer check weight $w = 6$.** The code rate is $\frac{k}{n}$, the efficiency ratio is $\frac{kd^2}{n}$, with $d = \min(d_X, d_Z)$; it is an upper bound when d is.

$[[n, k, (\leq d_X, \leq d_Z)]]$	(ℓ, m, p, q)	k/n	F	G	H	I
$[[144, 12, (8, 8)]]$	3, 2, 2, 2	0.083	$w^2xyz + wy + yz$	$w^2y + wxz + x$	$w^2x + wz + xy$	$w^2yz + w + 1$
$[[144, 12, (12, 12)]]$	3, 2, 2, 2	0.083	$w^2xy + w + xyz$	$w^2x + wxy + z$	$w^2xz + wx + xyz$	$w^2yz + wxy + xyz$
$[[216, 12, (12, 12)]]$	2, 2, 3, 3	0.056	$wxy^2 + wz^2 + x$	$wxyz^2 + wy^2z + x$	$wxy^2z^2 + xyz^2 + xz^2$	$wz + w + xy^2z$
$[[216, 24, (\leq 9, \leq 9)]]$	2, 2, 3, 3	0.111	$wy^2z^2 + x + yz^2$	$wxz^2 + xy^2z + y$	$wxy^2z + wxyz^2 + wxz^2$	$wyz + xy^2 + z$
$[[324, 12, (\leq 18, \leq 18)]]$	2, 3, 3, 3	0.037	$wx^2y + x^2y^2 + z$	$wy + xy + y^2z^2$	$wx^2z + wyz + yz^2$	$wx^2y^2 + wxyz^2 + xy^2z$
$[[336, 18, (\leq 12, \leq 12)]]$	2, 2, 2, 7	0.054	$wxyz^6 + wx + z^2$	$wz^3 + xyz^6 + z^4$	$xy + xz^5 + yz^4$	$wxyz^3 + wyz + wy$
$[[432, 12, (\leq 21, \leq 21)]]$	2, 3, 3, 4	0.028	$wx^2yz^3 + wx^2z + y^2$	$wxz^2 + x^2y + y^2z^2$	$wxz^2 + wy + xy^2z$	$wy + wz + x^2y^2z^3$
$[[432, 24, (\leq 18, \leq 18)]]$	2, 2, 3, 6	0.056	$wxz + wy^2z^4 + xy$	$wxy^2 + wxyz + wx$	$wxy^2z^4 + wyz + xz^2$	$wxy^2z^5 + wyz^4 + xz^4$
$[[486, 12, (\leq 18, \leq 18)]]$	3, 3, 3, 3	0.025	$wx^2z + wxz + x^2y^2z^2$	$w^2xy + w^2 + wx^2y$	$w^2xyz^2 + wxy + xz^2$	$w^2yz + w^2 + yz$
$[[486, 24, (12, 12)]]$	3, 3, 3, 3	0.049	$wyz + y^2z^2 + y^2$	$w^2z^2 + x^2y^2z^2 + z$	$w^2x^2z^2 + wx^2y^2z + wy^2z$	$w^2x^2y^2z + wz + x^2z^2$
$[[504, 12, (\leq 18, \leq 18)]]$	2, 2, 3, 7	0.024	$y^2z^4 + yz^5 + z$	$xy^2z^3 + xz^6 + yz^4$	$wxz + xy^2z^4 + yz^2$	$wz^3 + xy^2z^5 + xyz$

TABLE III (continued)

$[[n,k,(\leq d_X,\leq d_Z)]]$	(ℓ,m,p,q)	k/n	F	G	H	I
$[[504,12,(\leq 28,\leq 28)]]$	2, 2, 3, 7	0.024	wyz^2+y^2z $+z^4$	$wxy^2z^3+xz^5$ $+yz^6$	$wxyz^6+xy^2z^6$ $+xz^3$	$wxz^2+wy^2z^2$ $+yz^3$
$[[540,12,(\leq 31,\leq 31)]]$	2, 3, 3, 5	0.022	y^2z^4+yz+ z	$wx^2yz^4+x^2y^2z^3$ $+xy$	$wx^2y^2+xy^2z^2$ $+xyz^3$	wxz^3+x^2yz $+xy$
$[[576,12,(\leq 24,\leq 24)]]$	2, 2, 3, 8	0.021	wxz^7+y^2z $+yz^6$	$wxz^7+wy^2z^4$ $+yz^4$	$wx+wyz^3$ $+y^2z^3$	$wxy^2z^4+wxyz^2$ $+wxz^3$
$[[648,12,(\leq 22,\leq 22)]]$	3, 9, 2, 2	0.019	$w^2x^4yz+w^2x^2y$ $+wx^5yz$	w^2xz+wx^7yz $+wx^5yz$	wx^5yz+x^8y+y	$w^2x^8z+wx^8yz$ $+x^5$
$[[648,12,(\leq 24,\leq 24)]]$	3, 3, 3, 4	0.019	$w^2x^2z+w^2yz^2$ $+z^3$	$w^2x^2z^2+w^2xyz^2$ $+x^2z^2$	$w^2x^2y^2z^2+x^2yz$ $+y^2z$	$w^2x^2y^2+wx^2yz^3$ $+x^2yz^3$
$[[648,12,(\leq 31,\leq 31)]]$	3, 3, 3, 4	0.019	$w^2y^2z+wy^2$ $+yz$	$1+w^2xy^2z^2$ $+wx^2y^2z^3$	wx^2y+wxy^2z $+x^2y^2$	w^2yz^2+wxz $+wz$
$[[672,18,(\leq 20,\leq 20)]]$	2, 2, 2, 14	0.027	xz^5+xz^3 $+z^{13}$	wyz^6+wz^5 $+xyz^3$	wxz^6+z^{10} $+z^5$	$wxyz^{10}+wxz^4$ $+yz$
$[[720,12,(\leq 22,\leq 22)]]$	2, 2, 5, 6	0.017	wz^4+x $+y^2z^2$	$wxy^2z^2+wxz^3$ $+wyz$	$wy^4z^2+xy^3$ $+xy^2z^4$	$wxyz^5+xy^3$ $+yz^4$
$[[720,12,(\leq 40,\leq 40)]]$	2, 2, 2, 15	0.017	wxz^2+wz^3 $+z^4$	wyz^4+xz^5 $+yz^3$	$wz^{14}+x$ $+yz^4$	$wxyz^6+yz^{10}$ $+z^{14}$
$[[756,12,(\leq 22,\leq 22)]]$	2, 3, 3, 7	0.016	$wx^2z^3+wxy^2z^5$ $+xyz^6$	wy^2+wyz^3 $+x^2$	wy^2+wz^3 $+xyz^6$	$wxz^6+wy^2z^2$ $+x^2yz^2$
$[[756,12,(\leq 37,\leq 37)]]$	2, 3, 3, 7	0.016	$wxy^2z^4+x^2yz^2$ $+z^3$	$wz^4+x^2yz^5$ $+yz^6$	wy^2+wyz^6 $+z^5$	wx^2y+wyz^6 $+z^2$
$[[810,12,(\leq 38,\leq 38)]]$	3, 3, 3, 5	0.015	$w^2x^2y+wx^2yz^2$ $+z$	$w^2x^2z^4+wyz$ $+y^2z^2$	$w^2xyz^2+wyz^4$ $+y^2z^2$	$w^2x^2z^3+wxyz^2$ $+yz^3$
$[[840,18,(\leq 46,\leq 46)]]$	2, 2, 5, 7	0.021	$wxy^4z^3+wxy+z^2$	$wxyz+wy^2$ $+yz^5$	$wz^4+y^3z^3$ $+yz$	$wxy^3z^6+wxy^2z^2$ $+wy^4z$
$[[864,12,(\leq 18,\leq 18)]]$	2, 2, 3, 12	0.014	$wy^2z^5+xz^{11}$ $+z^3$	$wxz^{10}+wy^2z^4$ $+xz^5$	$wxyz^5+wyz^9$ $+yz^7$	wxz^4+wyz^{11} $+z^{11}$
$[[864,12,(\leq 48,\leq 48)]]$	2, 2, 3, 12	0.014	$wxyz^4+xyz^9$ $+y^2z^6$	$wy+xz^6$ $+z^7$	wyz^4+yz^6 $+yz^5$	$wxyz^{11}+wxz^2$ $+wx$
$[[972,12,(\leq 24,\leq 24)]]$	3, 3, 3, 6	0.012	$w^2x^2y^2z^5+wxy^2z^3$ $+xy^2z^4$	$w^2xz^3+wy^2$ $+xy^2z$	$w^2x^2+w^2y^2z^3$ $+w^2y$	$w^2x^2y^2z+w^2yz$ $+wx^2$
$[[972,24,(\leq 24,\leq 24)]]$	3, 3, 3, 6	0.025	$w^2x^2y^2z^5+w^2x^2z^3$ $+wz^3$	$w^2xyz+wx^2yz^4$ $+wx^2z^5$	$w^2x^2y^2z^4+wxyz$ $+yz^4$	$w^2x^2y+w^2x^2z^4$ $+wy^2z$
$[[972,24,(\leq 24,\leq 24)]]$	3, 3, 3, 6	0.025	wyz^2+wz $+z^2$	$w^2xy^2z+x^2yz^4$ $+yz^3$	$w^2xz^2+wx^2z$ $+wy^2z^4$	wz^5+wz $+xyz^3$

TABLE III: **Small instances of multivariate multicycle (MM) codes with stabilizer weight $w=9$.**

$[[n,k,(\leq d_X,\leq d_Z)]]$	(ℓ,m,p,q)	k/n	w^{med}	$w^{\max}$	F	G	H	I
$[[96,44,(4,4)]]$	2, 2, 2, 2	0.458	12	12	$(1+x)(1+yz)$	$(1+y)(1+zw)$	$(1+z)(1+wx)$	$(1+w)(1+xy)$
$[[144,12,(8,8)]]$	2, 2, 2, 3	0.083	9	10	$wxy+xyz$	$y+zx$ $+yx+zw$	$x+zyxw$	$z+y$ $+x+zyx$
$[[192,12,(\leq 12,\leq 12)]]$	2, 2, 2, 4	0.062	12	12	$wxyz^3+xyz$ $+xz^3+xz^2$	wxz^2+wyz^3 $+w+1$	$wy+wz^2$ $+xyz^2+xz$	$wxyz+wy$ $+xz^3+z^3$
$[[216,12,(12,12)]]$	2, 2, 3, 3	0.056	9	10	$wxy+xyz$	$y+zx$ $+yx+zw$	$x+zyxw$	$z+y$ $+x+zyx$
$[[216,12,(\leq 14,\leq 14)]]$	2, 2, 3, 3	0.056	12	12	wz^2+xy^2z $+xy^2+y$	$wxy^2z^2+wxz^2$ $+wy^2z+y^2z^2$	wxy^2z^2+wxyz $+xy^2z+x$	wxz^2+wyz $+wz^2+xz$
$[[240,6,(\leq 20,\leq 20)]]$	2, 2, 2, 5	0.025	12	12	wxz^4+wyz^3 $+wy+xyz^4$	wyz^4+wz $+w+xz^3$	$wxz+wyz^4$ $+wz^4+xz^4$	$wxyz^4+wxyz^3$ $+wyz+w$

TABLE IV (continued)

$[[n,k,(\leq d_X,\leq d_Z)]]$	(ℓ,m,p,q)	k/n	w^{med}	w^{max}	F	G	H	I
$[[240,12,(\leq 16,\leq 16)]]$	2, 2, 2, 5	0.050	12	12	$wxz+wz^4$ $+xyz^3+yz$	wz^3+xz^4 $+yz^2+z$	wz^2+xz $+yz^4+z$	xz^4+yz^3 $+z^3+z^2$
$[[240,18,(\leq 10,\leq 10)]]$	2, 2, 2, 5	0.075	12	12	$wxyz^3+wxz^2$ $+wz^3+z$	$wxyz^4+wxy$ $+wx+xy$	wyz^4+wyz^3 $+wz^2+yz^3$	$wxyz+wx$ $+wyz^4+wz$
$[[288,24,(\leq 12,\leq 12)]]$	2, 2, 2, 6	0.083	12	12	$wxyz^5+wyz^2$ $+xyz^4+y$	$wxyz^5+wxy$ $+wz+xyz$	$wxyz^4+wyz^5$ $+wyz^4+z^4$	wxz^4+wyz^4 $+wz^3+z^5$
$[[324,6,(\leq 18,\leq 18)]]$	2, 3, 3, 3	0.019	12	12	$wx^2z^2+x^2y^2$ $+x^2yz^2+y$	$wxz+wx$ $+x^2y^2z+xy^2z$	$wyz+x^2y^2z^2$ $+x^2z+y$	$wx^2yz^2+wxy^2z$ $+x^2y+y^2z^2$
$[[324,18,(\leq 18,\leq 18)]]$	2, 3, 3, 3	0.056	12	12	wx^2z+wy^2z $+wz^2+x^2yz^2$	$wx^2y^2z+wx^2yz^2$ $+wxy^2z^2+x^2z$	$wxy+wxz$ $+wx+y^2z$	$wx^2y^2z+wy^2$ $+wz+w$
$[[336,6,(\leq 28,\leq 28)]]$	2, 2, 2, 7	0.018	12	12	$wxyz^4+wz^4$ $+yz^6+yz^3$	wyz^6+w $+xz^3+yz^5$	wz^3+xyz^6 $+xz^3+xz$	wxz^4+wz^2 $+xyz^2+1$
$[[336,12,(\leq 24,\leq 24)]]$	2, 2, 2, 7	0.036	12	12	$wxyz^5+wxyz$ $+wyz^3+wz^4$	wxz^5+xy $+yz^6+z^4$	$wxyz^2+wz^5$ $+xz^3+yz^5$	$wxyz^3+wxz^3$ $+wyz^6+y$
$[[384,24,(\leq 16,\leq 16)]]$	2, 2, 4, 4	0.062	12	12	$wxyz^3+wy^2z^3$ $+xy^2+y^2z$	$wxyz^2+wxz^3$ $+wy^3z+xyz^3$	wxy^3z+wxz^2 $+wy^2z^3+xyz^2$	wxy^3z+wxz^3 $+xy^2z^2+xy^2z$
$[[432,12,(\leq 18,\leq 18)]]$	2, 2, 2, 9	0.028	12	12	$wxyz^6+wxyz$ $+y+z^8$	$wxyz^5+wyz^8$ $+xz^2+yz^3$	wxz^2+wz^3 $+xyz^2+yz^6$	wyz^3+xz^7 $+xz^2+x$
$[[480,12,(\leq 20,\leq 20)]]$	2, 2, 2, 10	0.025	12	12	$wxyz^9+wxz^7$ $+w+z$	wxz^4+wyz^5 $+xz+z^5$	xyz^6+xyz^5 $+xy+z^7$	$wxyz^8+wyz^9$ $+w+z^8$
$[[486,24,(12,12)]]$	3, 3, 3, 3	0.049	9	9	$1+wx$ $+x^2y$	$1+xy$ $+y^2z$	$1+yz$ $+wz^2$	$1+wz$ $+w^2x$
$[[486,66,(9,9)]]$	3, 3, 3, 3	0.136	12	12	$(1+x)(1+yz)$	$(1+y)(1+zw)$	$(1+z)(1+wx)$	$(1+w)(1+xy)$
$[[480,12,(\leq 20,\leq 20)]]$	2, 2, 2, 10	0.025	12	12	wxz^5+wz^7 $+xz^8+1$	$xy+yz^3$ $+yz^2+1$	$w+xyz^8$ $+yz^3+z^4$	wz^8+xz^9 $+xz^4+z^6$
$[[576,64,(6,6)]]$	2, 3, 4, 4	0.111	12	12	$(1+x)(1+yz)$	$(1+y)(1+zw)$	$(1+z)(1+wx)$	$(1+w)(1+xy)$
$[[624,12,(\leq 28,\leq 28)]]$	2, 2, 2, 13	0.019	12	12	$wxyz^{10}+wxz^6$ $+wz^4+z^{12}$	wxz^7+wxz^5 $+wz+yz^3$	$wxyz^4+xyz^6$ $+xz^{10}+yz^5$	$wxyz^2+xyz^{11}$ $+yz^7+z^2$
$[[648,60,(9,9)]]$	3, 3, 3, 4	0.093	12	12	$(1+x)(1+yz)$	$(1+y)(1+zw)$	$(1+z)(1+wx)$	$(1+w)(1+xy)$
$[[672,12,(\leq 28,\leq 28)]]$	2, 2, 4, 7	0.018	12	12	wxz^3+wyz^4 $+xy^3+z^4$	$wxy^2z^4+wxyz^2$ $+wz^3+yz$	$wxy^3z^4+wxy^2z^6$ $+wyz^6+yz^6$	$wxyz+x$ $+y^2z^5+yz^4$
$[[720,12,(\leq 27,\leq 27)]]$	2, 3, 4, 5	0.017	12	12	$wx^2z^4+wyz^3$ $+x^2y^3z^2+y^3$	$wx^2yz^4+wx^2y$ $+wxyz^2+wy^2z^3$	$wx^2y^3z+wx^2z^3$ $+x^2y^3z+x^2y^2$	$wxy^2z^3+wxyz^2$ $+x^2y+xy^2z^4$
$[[768,12,(\leq 30,\leq 30)]]$	2, 2, 2, 16	0.016	12	12	$wyz^{13}+wyz^{12}$ $+xyz^{11}+z^{15}$	wxz^2+wz^6+ $xyz^{11}+yz^{12}$	$wxyz^{11}+xyz^{10}$ $+xz^7+z^{14}$	$wxy+wxz^9$ $+wy+yz$

TABLE IV: **Parameters and explicit instances of MM codes with metachecks for both types of checks.** Here, n is the number of physical qubits, k the number of logical qubits, the pairs give the X-type and Z-type distances or upper bounds on them, and w^{med} and w^{max} are the median and maximum row weights of the stabilizer checks. The multivariate quotient ring polynomials $F(w,x,y,z)$, $G(w,x,y,z)$, $H(w,x,y,z)$, and $I(w,x,y,z)$ are defined over $\mathbb{F}_2[w,x,y,z]/\langle w^\ell-1, x^m-1, y^p-1, z^r-1\rangle$.

$[[n,k,(\leq d_X,\leq d_Z)]]$	(ℓ,m,p,q)	k/n	F	G	H	I
$[[96,12,(8,8)]]$	2, 2, 2, 2	0.125	$wy+wz$ $+w+1$	$wxz+wy$ $+wz+w$	$wxyz+w$ $+xyz+y$	$wxyz+w$ $+yz+1$
$[[144,6,(8,8)]]$	3, 2, 2, 2	0.042	$wxyz+wxy$ $+wy+yz$	w^2z+xyz $+xy+yz$	w^2x+w^2yz $+w^2+1$	w^2z+wy $+w+xz$
$[[144,6,(10,10)]]$	3, 2, 2, 2	0.042	w^2xz+w^2yz $+wx+z$	$w^2yz+wxz$ $+w+xy$	$wxy+wy$ $+x+y$	$wxz+xy$ $+x+z$

TABLE V (continued)

$[[n,k,(\leq d_X,\leq d_Z)]]$	(ℓ,m,p,q)	k/n	F	G	H	I
$[[144,6,(\leq 12,\leq 12)]]$	3, 2, 2, 2	0.042	w^2+wyz $+wz+w$	w^2z+wxy $+yz+z$	w^2y+wz $+w+1$	w^2z+wx $+wy+1$
$[[144,12,(8,8)]]$	2, 3, 2, 2	0.083	x^2z+x^2 $+yz+z$	wx^2y+w $+x^2+yz$	$wxy+w$ $+x^2y+z$	$wxyz+wxy+x^2y+xz$
$[[144,12,(10,10)]]$	3, 2, 2, 2	0.083	w^2x+xyz $+x+y$	w^2y+wxy $+wx+z$	w^2yz+w^2 $+yz+z$	$w^2x+wxyz+wyz+1$
$[[144,12,(\leq 12,\leq 12)]]$	3, 2, 2, 2	0.083	w^2xz+w^2yz $+wy+xy$	$wxy+wyz$ $+wz+xy$	w^2xy+w^2 $+wyz+1$	$w^2xz+w^2z+wxz+xy$

TABLE V: **Multivariate multicycle (MM) codes with stabilizer weight $w=12$.**

$[[n,k,(\leq d_X,\leq d_Z)]]$	ℓ	k/n	F	G	H	I
$[[162,6,(9,9)]]$	27	0.037	$1+w^7$	$1+w^{13}$	$1+w^{15}$	$1+w^{24}$
$[[168,6,(9,9)]]$	28	0.036	$1+w^4$	$1+w^7$	$1+w^{13}$	$1+w^{19}$
$[[168,12,(7,7)]]$	28	0.071	$1+w^4$	$1+w^6$	$1+w^8$	$1+w^{18}$
$[[174,6,(11,11)]]$	29	0.034	$1+w^2$	$1+w^8$	$1+w^{20}$	$1+w^{24}$
$[[180,6,(10,10)]]$	30	0.033	$1+w^3$	$1+w^{13}$	$1+w^{21}$	$1+w^{28}$
$[[186,6,(11,11)]]$	31	0.032	$1+w$	$1+w^{14}$	$1+w^{19}$	$1+w^{23}$
$[[198,6,(10,10)]]$	33	0.030	$1+w^3$	$1+w^7$	$1+w^{16}$	$1+w^{21}$
$[[204,6,(\leq 12,\leq 12)]]$	34	0.029	$1+w^5$	$1+w^8$	$1+w^9$	$1+w^{23}$
$[[210,6,(\leq 11,\leq 11)]]$	35	0.029	$1+w^{13}$	$1+w^{25}$	$1+w^{29}$	$1+w^{34}$
$[[216,6,(\leq 12,\leq 12)]]$	36	0.028	$1+w^3$	$1+w^{21}$	$1+w^{26}$	$1+w^{35}$
$[[222,6,(\leq 11,\leq 11)]]$	37	0.027	$1+w^4$	$1+w^{17}$	$1+w^{22}$	$1+w^{34}$
$[[228,6,(\leq 12,\leq 12)]]$	38	0.026	$1+w^3$	$1+w^{24}$	$1+w^{30}$	$1+w^{37}$
$[[234,6,(\leq 13,\leq 13)]]$	39	0.026	$1+w$	$1+w^{14}$	$1+w^{30}$	$1+w^{36}$
$[[240,6,(\leq 12,\leq 12)]]$	40	0.025	$1+w^2$	$1+w^5$	$1+w^{11}$	$1+w^{26}$
$[[246,6,(\leq 12,\leq 12)]]$	41	0.024	$1+w$	$1+w^{12}$	$1+w^{23}$	$1+w^{37}$
$[[252,6,(\leq 12,\leq 12)]]$	42	0.024	$1+w^9$	$1+w^{17}$	$1+w^{19}$	$1+w^{30}$
$[[258,6,(\leq 12,\leq 12)]]$	43	0.023	$1+w^{15}$	$1+w^{16}$	$1+w^{34}$	$1+w^{39}$
$[[264,6,(\leq 13,\leq 13)]]$	44	0.023	$1+w^6$	$1+w^7$	$1+w^{21}$	$1+w^{26}$
$[[270,6,(\leq 14,\leq 14)]]$	45	0.022	$1+w^9$	$1+w^{24}$	$1+w^{32}$	$1+w^{38}$
$[[276,6,(\leq 14,\leq 14)]]$	46	0.022	$1+w^{19}$	$1+w^{32}$	$1+w^{34}$	$1+w^{43}$
$[[282,6,(\leq 13,\leq 13)]]$	47	0.021	$1+w^{23}$	$1+w^{25}$	$1+w^{28}$	$1+w^{39}$
$[[288,6,(\leq 13,\leq 13)]]$	48	0.021	$1+w^{22}$	$1+w^{23}$	$1+w^{41}$	$1+w^{43}$
$[[294,6,(\leq 14,\leq 14)]]$	49	0.020	$1+w^9$	$1+w^{15}$	$1+w^{22}$	$1+w^{48}$
$[[300,6,(\leq 13,\leq 13)]]$	50	0.020	$1+w^{12}$	$1+w^{16}$	$1+w^{21}$	$1+w^{23}$
$[[306,6,(\leq 14,\leq 14)]]$	51	0.020	$1+w$	$1+w^{36}$	$1+w^{39}$	$1+w^{46}$
$[[312,6,(\leq 13,\leq 13)]]$	52	0.019	$1+w^8$	$1+w^{17}$	$1+w^{41}$	$1+w^{42}$
$[[318,6,(\leq 13,\leq 13)]]$	53	0.019	$1+w$	$1+w^{19}$	$1+w^{21}$	$1+w^{29}$
$[[324,6,(\leq 14,\leq 14)]]$	54	0.019	$1+w^9$	$1+w^{11}$	$1+w^{44}$	$1+w^{48}$
$[[330,6,(\leq 14,\leq 14)]]$	55	0.018	$1+w^{15}$	$1+w^{39}$	$1+w^{46}$	$1+w^{51}$
$[[342,6,(\leq 14,\leq 14)]]$	57	0.018	$1+w$	$1+w^{15}$	$1+w^{46}$	$1+w^{49}$
$[[348,6,(\leq 15,\leq 15)]]$	58	0.017	$1+w^{22}$	$1+w^{43}$	$1+w^{55}$	$1+w^{56}$

TABLE VI (continued)

$[[n,k,(\leq d_X,\leq d_Z)]]$	ℓ	k/n	F	G	H	I
$[[354,6,(\leq 15,\leq 15)]]$	59	0.017	$1+w^{33}$	$1+w^{39}$	$1+w^{42}$	$1+w^{47}$
$[[360,6,(\leq 15,\leq 15)]]$	60	0.017	$1+w^{4}$	$1+w^{18}$	$1+w^{25}$	$1+w^{44}$
$[[366,6,(\leq 15,\leq 15)]]$	61	0.016	$1+w^{30}$	$1+w^{38}$	$1+w^{47}$	$1+w^{48}$
$[[372,6,(\leq 16,\leq 16)]]$	62	0.016	$1+w^{42}$	$1+w^{46}$	$1+w^{55}$	$1+w^{57}$
$[[378,6,(\leq 17,\leq 17)]]$	63	0.016	$1+w^{11}$	$1+w^{44}$	$1+w^{46}$	$1+w^{56}$
$[[390,6,(\leq 15,\leq 15)]]$	65	0.015	$1+w^{8}$	$1+w^{15}$	$1+w^{39}$	$1+w^{59}$
$[[396,6,(\leq 15,\leq 15)]]$	66	0.015	$1+w^{12}$	$1+w^{13}$	$1+w^{32}$	$1+w^{38}$
$[[402,6,(\leq 16,\leq 16)]]$	67	0.015	$1+w^{7}$	$1+w^{9}$	$1+w^{45}$	$1+w^{57}$
$[[408,6,(\leq 15,\leq 15)]]$	68	0.015	$1+w^{4}$	$1+w^{46}$	$1+w^{49}$	$1+w^{52}$
$[[414,6,(\leq 16,\leq 16)]]$	69	0.014	$1+w^{9}$	$1+w^{19}$	$1+w^{24}$	$1+w^{53}$
$[[420,6,(\leq 15,\leq 15)]]$	70	0.014	$1+w^{2}$	$1+w^{13}$	$1+w^{32}$	$1+w^{56}$
$[[426,6,(\leq 17,\leq 17)]]$	71	0.014	$1+w^{15}$	$1+w^{19}$	$1+w^{29}$	$1+w^{35}$
$[[432,6,(\leq 16,\leq 16)]]$	72	0.014	$1+w^{5}$	$1+w^{13}$	$1+w^{50}$	$1+w^{60}$
$[[438,6,(\leq 17,\leq 17)]]$	73	0.014	$1+w^{18}$	$1+w^{29}$	$1+w^{32}$	$1+w^{63}$
$[[444,6,(\leq 17,\leq 17)]]$	74	0.014	$1+w^{6}$	$1+w^{26}$	$1+w^{49}$	$1+w^{59}$
$[[450,6,(\leq 16,\leq 16)]]$	75	0.013	$1+w^{5}$	$1+w^{49}$	$1+w^{63}$	$1+w^{64}$
$[[456,6,(\leq 16,\leq 16)]]$	76	0.013	$1+w^{17}$	$1+w^{28}$	$1+w^{68}$	$1+w^{74}$
$[[462,6,(\leq 17,\leq 17)]]$	77	0.013	$1+w^{9}$	$1+w^{26}$	$1+w^{37}$	$1+w^{53}$
$[[468,6,(\leq 17,\leq 17)]]$	78	0.013	$1+w^{24}$	$1+w^{33}$	$1+w^{38}$	$1+w^{65}$
$[[474,6,(\leq 17,\leq 17)]]$	79	0.013	$1+w^{7}$	$1+w^{10}$	$1+w^{12}$	$1+w^{61}$
$[[480,6,(\leq 16,\leq 16)]]$	80	0.013	$1+w^{7}$	$1+w^{13}$	$1+w^{35}$	$1+w^{44}$
$[[486,6,(\leq 17,\leq 17)]]$	81	0.012	$1+w$	$1+w^{14}$	$1+w^{21}$	$1+w^{64}$
$[[492,6,(\leq 16,\leq 16)]]$	82	0.012	$1+w^{8}$	$1+w^{21}$	$1+w^{27}$	$1+w^{38}$
$[[498,6,(\leq 18,\leq 18)]]$	83	0.012	$1+w^{7}$	$1+w^{9}$	$1+w^{31}$	$1+w^{79}$
$[[504,6,(\leq 18,\leq 18)]]$	84	0.012	$1+w^{15}$	$1+w^{45}$	$1+w^{68}$	$1+w^{73}$
$[[510,6,(\leq 17,\leq 17)]]$	85	0.012	$1+w^{5}$	$1+w^{22}$	$1+w^{55}$	$1+w^{84}$
$[[516,6,(\leq 18,\leq 18)]]$	86	0.012	$1+w^{5}$	$1+w^{13}$	$1+w^{62}$	$1+w^{69}$
$[[522,6,(\leq 20,\leq 20)]]$	87	0.011	$1+w^{39}$	$1+w^{54}$	$1+w^{77}$	$1+w^{80}$

TABLE VI: **Collapsed 4-dimensional AMC codes via the Multivariate Multicycle (MM) framework.** These codes include instances related to Lin *et al.*, arXiv:2506.16910; the distances given here are from our own search. The multivariate quotient ring polynomials $F(w,x,y,z)$, $G(w,x,y,z)$, $H(w,x,y,z)$, and $I(w,x,y,z)$ are defined over $\mathbb{F}_2[w,x,y,z]/\langle w^\ell-1, x-1, y-1, z-1\rangle$. All these codes have weight-6 X and Z stabilizer checks.

$[[n,k,(\leq d_X,\leq d_Z)]]$	ℓ	k/n	F	G	H	I	J
$[[180,10,(12,7)]]$	18	0.056	$1+v$	$1+v^{5}$	$1+v^{10}$	$1+v^{11}$	$1+v^{14}$
$[[190,10,(12,7)]]$	19	0.053	$1+v$	$1+v^{4}$	$1+v^{5}$	$1+v^{11}$	$1+v^{17}$
$[[190,10,(13,7)]]$	19	0.053	$1+v^{4}$	$1+v^{10}$	$1+v^{12}$	$1+v^{16}$	$1+v^{18}$
$[[200,10,(\leq 12,\leq 7)]]$	20	0.050	$1+v$	$1+v^{2}$	$1+v^{9}$	$1+v^{14}$	$1+v^{15}$
$[[210,10,(\leq 14,\leq 7)]]$	21	0.048	$1+v^{3}$	$1+v^{4}$	$1+v^{8}$	$1+v^{10}$	$1+v^{15}$

TABLE VII (continued)

$[[n,k,(\leq d_X,\leq d_Z)]]$	ℓ	k/n	F	G	H	I	J
$[[220,10,(\leq 14,\leq 7)]]$	22	0.045	$1+v^4$	$1+v^7$	$1+v^{10}$	$1+v^{19}$	$1+v^{21}$
$[[230,10,(\leq 13,\leq 7)]]$	23	0.043	$1+v^3$	$1+v^5$	$1+v^{12}$	$1+v^{16}$	$1+v^{19}$
$[[240,10,(\leq 13,\leq 7)]]$	24	0.042	$1+v^4$	$1+v^7$	$1+v^{13}$	$1+v^{21}$	$1+v^{23}$
$[[240,10,(\leq 14,\leq 7)]]$	24	0.042	$1+v$	$1+v^9$	$1+v^{10}$	$1+v^{17}$	$1+v^{20}$
$[[250,10,(\leq 13,\leq 7)]]$	25	0.040	$1+v^8$	$1+v^{11}$	$1+v^{12}$	$1+v^{15}$	$1+v^{23}$
$[[250,10,(\leq 14,\leq 7)]]$	25	0.040	$1+v^4$	$1+v^5$	$1+v^6$	$1+v^9$	$1+v^{14}$
$[[250,10,(\leq 15,\leq 8)]]$	25	0.040	$1+v$	$1+v^4$	$1+v^9$	$1+v^{15}$	$1+v^{22}$
$[[250,10,(\leq 16,\leq 8)]]$	25	0.040	$1+v^4$	$1+v^9$	$1+v^{13}$	$1+v^{14}$	$1+v^{19}$
$[[260,10,(\leq 13,\leq 7)]]$	26	0.038	$1+v^5$	$1+v^6$	$1+v^9$	$1+v^{12}$	$1+v^{24}$
$[[260,10,(\leq 14,\leq 7)]]$	26	0.038	$1+v^8$	$1+v^{15}$	$1+v^{20}$	$1+v^{23}$	$1+v^{25}$
$[[260,10,(\leq 15,\leq 8)]]$	26	0.038	$1+v^3$	$1+v^5$	$1+v^6$	$1+v^7$	$1+v^{11}$
$[[270,10,(\leq 13,\leq 7)]]$	27	0.037	$1+v^7$	$1+v^{10}$	$1+v^{11}$	$1+v^{12}$	$1+v^{23}$
$[[270,10,(\leq 16,\leq 7)]]$	27	0.037	$1+v^5$	$1+v^6$	$1+v^{17}$	$1+v^{18}$	$1+v^{20}$
$[[280,10,(\leq 15,\leq 7)]]$	28	0.036	$1+v^3$	$1+v^5$	$1+v^{15}$	$1+v^{16}$	$1+v^{26}$
$[[290,10,(\leq 15,\leq 8)]]$	29	0.034	$1+v$	$1+v^3$	$1+v^5$	$1+v^6$	$1+v^{15}$
$[[300,10,(\leq 15,\leq 9)]]$	30	0.033	$1+v^4$	$1+v^5$	$1+v^{18}$	$1+v^{19}$	$1+v^{28}$
$[[310,10,(\leq 15,\leq 8)]]$	31	0.032	$1+v^3$	$1+v^{15}$	$1+v^{18}$	$1+v^{22}$	$1+v^{23}$
$[[330,10,(\leq 18,\leq 9)]]$	33	0.030	$1+v^2$	$1+v^7$	$1+v^8$	$1+v^{11}$	$1+v^{12}$
$[[340,10,(\leq 20,\leq 9)]]$	34	0.029	$1+v^3$	$1+v^7$	$1+v^{18}$	$1+v^{19}$	$1+v^{29}$
$[[350,10,(\leq 18,\leq 9)]]$	35	0.029	$1+v^3$	$1+v^5$	$1+v^9$	$1+v^{19}$	$1+v^{34}$
$[[360,10,(\leq 20,\leq 8)]]$	36	0.028	$1+v^8$	$1+v^9$	$1+v^{15}$	$1+v^{19}$	$1+v^{31}$
$[[370,10,(\leq 16,\leq 8)]]$	37	0.027	$1+v^2$	$1+v^4$	$1+v^{21}$	$1+v^{27}$	$1+v^{28}$
$[[380,10,(\leq 20,\leq 8)]]$	38	0.026	$1+v^2$	$1+v^9$	$1+v^{16}$	$1+v^{17}$	$1+v^{35}$
$[[390,10,(\leq 21,\leq 9)]]$	39	0.026	$1+v^5$	$1+v^{15}$	$1+v^{21}$	$1+v^{23}$	$1+v^{35}$
$[[400,10,(\leq 21,\leq 9)]]$	40	0.025	$1+v^{18}$	$1+v^{28}$	$1+v^{31}$	$1+v^{33}$	$1+v^{39}$
$[[410,10,(\leq 21,\leq 10)]]$	41	0.024	$1+v^7$	$1+v^{14}$	$1+v^{25}$	$1+v^{26}$	$1+v^{38}$
$[[420,10,(\leq 22,\leq 9)]]$	42	0.024	$1+v^{15}$	$1+v^{23}$	$1+v^{29}$	$1+v^{31}$	$1+v^{37}$
$[[430,10,(\leq 22,\leq 10)]]$	43	0.023	$1+v^9$	$1+v^{10}$	$1+v^{13}$	$1+v^{16}$	$1+v^{28}$
$[[440,10,(\leq 22,\leq 10)]]$	44	0.023	$1+v$	$1+v^{10}$	$1+v^{16}$	$1+v^{21}$	$1+v^{25}$
$[[450,10,(\leq 20,\leq 9)]]$	45	0.022	$1+v^4$	$1+v^{10}$	$1+v^{84}$	$1+v^{21}$	$1+v^{26}$
$[[460,10,(\leq 22,\leq 10)]]$	46	0.022	$1+v^9$	$1+v^{16}$	$1+v^{20}$	$1+v^{42}$	$1+v^{43}$
$[[470,10,(\leq 21,\leq 9)]]$	47	0.021	$1+v^{17}$	$1+v^{38}$	$1+v^{39}$	$1+v^{40}$	$1+v^{44}$
$[[480,10,(\leq 24,\leq 10)]]$	48	0.021	$1+v^8$	$1+v^{11}$	$1+v^{15}$	$1+v^{17}$	$1+v^{38}$
$[[490,10,(\leq 24,\leq 10)]]$	49	0.020	$1+v^8$	$1+v^{10}$	$1+v^{21}$	$1+v^{34}$	$1+v^{45}$
$[[500,10,(\leq 23,\leq 11)]]$	50	0.020	$1+v^3$	$1+v^{21}$	$1+v^{36}$	$1+v^{41}$	$1+v^{49}$
$[[510,10,(\leq 25,\leq 11)]]$	51	0.020	$1+v^8$	$1+v^{18}$	$1+v^{20}$	$1+v^{27}$	$1+v^{37}$
$[[520,10,(\leq 24,\leq 10)]]$	52	0.019	$1+v^5$	$1+v^{23}$	$1+v^{39}$	$1+v^{45}$	$1+v^{51}$
$[[530,10,(\leq 13,\leq 7)]]$	53	0.019	$1+v^8$	$1+v^{21}$	$1+v^{29}$	$1+v^{39}$	$1+v^{40}$
$[[540,10,(\leq 25,\leq 12)]]$	54	0.019	$1+v$	$1+v^8$	$1+v^{37}$	$1+v^{41}$	$1+v^{51}$
$[[550,10,(\leq 26,\leq 10)]]$	55	0.018	$1+v^{26}$	$1+v^{35}$	$1+v^{37}$	$1+v^{39}$	$1+v^{41}$
$[[560,10,(\leq 25,\leq 11)]]$	56	0.018	$1+v^{30}$	$1+v^{35}$	$1+v^{46}$	$1+v^{48}$	$1+v^{53}$

TABLE VII (continued)

$[[n,k,(\leq d_X,\leq d_Z)]]$	ℓ	k/n	F	G	H	I	J
$[[570,10,(\leq 28,\leq 11)]]$	57	0.018	$1+v^8$	$1+v^{17}$	$1+v^{30}$	$1+v^{45}$	$1+v^{56}$
$[[580,10,(\leq 28,\leq 12)]]$	58	0.017	$1+v^{21}$	$1+v^{32}$	$1+v^{39}$	$1+v^{43}$	$1+v^{46}$
$[[590,10,(\leq 25,\leq 11)]]$	59	0.017	$1+v^5$	$1+v^6$	$1+v^9$	$1+v^{31}$	$1+v^{38}$
$[[600,10,(\leq 24,\leq 10)]]$	60	0.017	$1+v^5$	$1+v^{12}$	$1+v^{21}$	$1+v^{35}$	$1+v^{59}$
$[[610,10,(\leq 26,\leq 11)]]$	61	0.016	$1+v^{23}$	$1+v^{32}$	$1+v^{39}$	$1+v^{43}$	$1+v^{49}$
$[[620,10,(\leq 26,\leq 11)]]$	62	0.016	$1+v^9$	$1+v^{13}$	$1+v^{35}$	$1+v^{42}$	$1+v^{50}$
$[[630,10,(\leq 28,\leq 12)]]$	63	0.016	$1+v$	$1+v^{16}$	$1+v^{36}$	$1+v^{44}$	$1+v^{50}$
$[[650,10,(\leq 28,\leq 10)]]$	65	0.015	$1+v^{18}$	$1+v^{41}$	$1+v^{55}$	$1+v^{61}$	$1+v^{64}$
$[[660,10,(\leq 30,\leq 11)]]$	66	0.015	$1+v^{36}$	$1+v^{38}$	$1+v^{49}$	$1+v^{50}$	$1+v^{57}$
$[[670,10,(\leq 30,\leq 12)]]$	67	0.015	$1+v^4$	$1+v^{44}$	$1+v^{51}$	$1+v^{53}$	$1+v^{66}$
$[[680,10,(\leq 30,\leq 11)]]$	68	0.015	$1+v^{11}$	$1+v^{14}$	$1+v^{16}$	$1+v^{61}$	$1+v^{67}$
$[[690,10,(\leq 23,\leq 12)]]$	69	0.014	$1+v^3$	$1+v^9$	$1+v^{25}$	$1+v^{42}$	$1+v^{65}$
$[[700,10,(\leq 27,\leq 11)]]$	70	0.014	$1+v^{26}$	$1+v^{36}$	$1+v^{41}$	$1+v^{46}$	$1+v^{59}$
$[[710,10,(\leq 30,\leq 13)]]$	71	0.014	$1+v^7$	$1+v^{18}$	$1+v^{24}$	$1+v^{45}$	$1+v^{61}$
$[[720,10,(\leq 30,\leq 12)]]$	72	0.014	$1+v^9$	$1+v^{28}$	$1+v^{33}$	$1+v^{47}$	$1+v^{68}$
$[[730,10,(\leq 29,\leq 12)]]$	73	0.014	$1+v^{43}$	$1+v^{52}$	$1+v^{56}$	$1+v^{59}$	$1+v^{71}$
$[[740,10,(\leq 28,\leq 10)]]$	74	0.014	$1+v^6$	$1+v^{18}$	$1+v^{21}$	$1+v^{43}$	$1+v^{54}$
$[[750,10,(\leq 32,\leq 13)]]$	75	0.013	$1+v^5$	$1+v^{22}$	$1+v^{41}$	$1+v^{59}$	$1+v^{62}$
$[[760,10,(\leq 24,\leq 8)]]$	76	0.013	$1+v^{19}$	$1+v^{29}$	$1+v^{44}$	$1+v^{67}$	$1+v^{74}$
$[[770,10,(\leq 32,\leq 13)]]$	77	0.013	$1+v^{21}$	$1+v^{40}$	$1+v^{48}$	$1+v^{62}$	$1+v^{65}$
$[[780,10,(\leq 26,\leq 11)]]$	78	0.013	$1+v^{43}$	$1+v^{48}$	$1+v^{50}$	$1+v^{59}$	$1+v^{72}$
$[[790,10,(\leq 34,\leq 12)]]$	79	0.013	$1+v^{12}$	$1+v^{22}$	$1+v^{23}$	$1+v^{42}$	$1+v^{63}$
$[[800,10,(\leq 32,\leq 12)]]$	80	0.013	$1+v^9$	$1+v^{15}$	$1+v^{17}$	$1+v^{20}$	$1+v^{47}$
$[[810,10,(\leq 27,\leq 12)]]$	81	0.012	$1+v^6$	$1+v^{34}$	$1+v^{48}$	$1+v^{55}$	$1+v^{63}$
$[[820,10,(\leq 31,\leq 11)]]$	82	0.012	$1+v^3$	$1+v^7$	$1+v^{40}$	$1+v^{66}$	$1+v^{77}$
$[[830,10,(\leq 28,\leq 13)]]$	83	0.012	$1+v^{39}$	$1+v^{47}$	$1+v^{55}$	$1+v^{70}$	$1+v^{82}$
$[[840,10,(\leq 29,\leq 12)]]$	84	0.012	$1+v^{18}$	$1+v^{29}$	$1+v^{61}$	$1+v^{68}$	$1+v^{69}$
$[[850,10,(\leq 35,\leq 14)]]$	85	0.012	$1+v$	$1+v^{12}$	$1+v^{31}$	$1+v^{34}$	$1+v^{47}$
$[[860,10,(\leq 28,\leq 12)]]$	86	0.012	$1+v^{20}$	$1+v^{23}$	$1+v^{49}$	$1+v^{55}$	$1+v^{64}$
$[[870,10,(\leq 33,\leq 14)]]$	87	0.011	$1+v^5$	$1+v^{18}$	$1+v^{26}$	$1+v^{72}$	$1+v^{73}$
$[[880,10,(\leq 34,\leq 13)]]$	88	0.011	$1+v^{16}$	$1+v^{53}$	$1+v^{57}$	$1+v^{63}$	$1+v^{65}$
$[[890,10,(\leq 34,\leq 14)]]$	89	0.011	$1+v^{10}$	$1+v^{19}$	$1+v^{66}$	$1+v^{72}$	$1+v^{78}$
$[[900,10,(\leq 30,\leq 13)]]$	90	0.011	$1+v^6$	$1+v^9$	$1+v^{39}$	$1+v^{40}$	$1+v^{58}$
$[[910,10,(\leq 38,\leq 16)]]$	91	0.011	$1+v^{24}$	$1+v^{35}$	$1+v^{47}$	$1+v^{73}$	$1+v^{81}$
$[[920,10,(\leq 34,\leq 12)]]$	92	0.011	$1+v^{33}$	$1+v^{35}$	$1+v^{37}$	$1+v^{49}$	$1+v^{52}$
$[[930,10,(\leq 33,\leq 13)]]$	93	0.011	$1+v^7$	$1+v^{10}$	$1+v^{16}$	$1+v^{64}$	$1+v^{72}$

TABLE VII: **Collapsed 5-dimensional (5D) multivariate multicycle (MM) codes.** The multivariate quotient ring polynomials $F(v,w,x,y,z)$, $G(v,w,x,y,z)$, $H(v,w,x,y,z)$, $I(v,w,x,y,z)$ and $J(v,w,x,y,z)$ are defined over $\mathbb{F}_2[v,w,x,y,z]/\langle v^\ell-1, w-1, x-1, y-1, z-1\rangle$. Notably, all these 5D MM codes have weight-8 X stabilizer checks and weight-6 Z stabilizer checks.

$[[n,k,(\leq d_X,\leq d_Z)]]$	ℓ	k/n	F	G	H	I	J	K
$[[200,20,(\leq 7,\leq 7)]]$	10	0.100	$1+u$	$1+u^3$	$1+u^6$	$1+u^7$	$1+u^8$	$1+u^9$
$[[220,20,(\leq 8,\leq 8)]]$	11	0.091	$1+u$	$1+u^3$	$1+u^5$	$1+u^7$	$1+u^8$	$1+u^{10}$
$[[240,20,(\leq 8,\leq 8)]]$	12	0.083	$1+u$	$1+u^2$	$1+u^3$	$1+u^5$	$1+u^9$	$1+u^{10}$
$[[260,20,(\leq 10,\leq 10)]]$	13	0.077	$1+u^5$	$1+u^7$	$1+u^9$	$1+u^{10}$	$1+u^{11}$	$1+u^{12}$
$[[280,20,(\leq 9,\leq 9)]]$	14	0.071	$1+u$	$1+u^2$	$1+u^3$	$1+u^4$	$1+u^5$	$1+u^{11}$
$[[300,20,(\leq 9,\leq 9)]]$	15	0.067	$1+u$	$1+u^3$	$1+u^5$	$1+u^6$	$1+u^8$	$1+u^{11}$
$[[320,20,(\leq 9,\leq 9)]]$	16	0.062	$1+u$	$1+u^3$	$1+u^5$	$1+u^7$	$1+u^{10}$	$1+u^{15}$
$[[340,20,(\leq 10,\leq 10)]]$	17	0.059	$1+u^5$	$1+u^7$	$1+u^9$	$1+u^{11}$	$1+u^{14}$	$1+u^{15}$
$[[360,20,(\leq 10,\leq 10)]]$	18	0.056	$1+u^3$	$1+u^7$	$1+u^{11}$	$1+u^{13}$	$1+u^{16}$	$1+u^{17}$
$[[380,20,(\leq 10,\leq 10)]]$	19	0.053	$1+u^2$	$1+u^3$	$1+u^5$	$1+u^6$	$1+u^{11}$	$1+u^{12}$
$[[380,20,(\leq 11,\leq 11)]]$	19	0.053	$1+u^9$	$1+u^{13}$	$1+u^{14}$	$1+u^{15}$	$1+u^{16}$	$1+u^{17}$
$[[400,20,(\leq 10,\leq 10)]]$	20	0.050	$1+u^8$	$1+u^{11}$	$1+u^{14}$	$1+u^{15}$	$1+u^{17}$	$1+u^{18}$
$[[400,20,(\leq 12,\leq 12)]]$	20	0.050	$1+u^3$	$1+u^7$	$1+u^{11}$	$1+u^{14}$	$1+u^{16}$	$1+u^{19}$
$[[400,20,(\leq 11,\leq 11)]]$	20	0.050	$1+u$	$1+u^2$	$1+u^3$	$1+u^6$	$1+u^9$	$1+u^{15}$
$[[420,20,(\leq 12,\leq 12)]]$	21	0.048	$1+u^7$	$1+u^{11}$	$1+u^{13}$	$1+u^{15}$	$1+u^{18}$	$1+u^{19}$
$[[440,20,(\leq 11,\leq 11)]]$	22	0.045	$1+u^7$	$1+u^9$	$1+u^{12}$	$1+u^{17}$	$1+u^{18}$	$1+u^{19}$
$[[460,20,(\leq 12,\leq 12)]]$	23	0.043	$1+u^4$	$1+u^7$	$1+u^{12}$	$1+u^{14}$	$1+u^{18}$	$1+u^{20}$
$[[480,20,(\leq 12,\leq 12)]]$	24	0.042	$1+u^{15}$	$1+u^{16}$	$1+u^{19}$	$1+u^{20}$	$1+u^{22}$	$1+u^{23}$
$[[520,20,(\leq 14,\leq 14)]]$	26	0.038	$1+u^3$	$1+u^5$	$1+u^7$	$1+u^{16}$	$1+u^{17}$	$1+u^{22}$
$[[540,20,(\leq 13,\leq 13)]]$	27	0.037	$1+u^3$	$1+u^4$	$1+u^{14}$	$1+u^{19}$	$1+u^{22}$	$1+u^{25}$
$[[560,20,(\leq 14,\leq 14)]]$	28	0.036	$1+u$	$1+u^6$	$1+u^{12}$	$1+u^{13}$	$1+u^{23}$	$1+u^{24}$
$[[580,20,(\leq 14,\leq 14)]]$	29	0.034	$1+u^3$	$1+u^5$	$1+u^9$	$1+u^{12}$	$1+u^{18}$	$1+u^{25}$
$[[600,20,(\leq 14,\leq 14)]]$	30	0.033	$1+u^{11}$	$1+u^{12}$	$1+u^{16}$	$1+u^{17}$	$1+u^{21}$	$1+u^{29}$
$[[620,20,(\leq 15,\leq 15)]]$	31	0.032	$1+u^4$	$1+u^5$	$1+u^{14}$	$1+u^{16}$	$1+u^{19}$	$1+u^{28}$
$[[640,20,(\leq 15,\leq 15)]]$	32	0.031	$1+u$	$1+u^2$	$1+u^5$	$1+u^7$	$1+u^{11}$	$1+u^{13}$
$[[660,20,(\leq 15,\leq 15)]]$	33	0.030	$1+u^5$	$1+u^8$	$1+u^{13}$	$1+u^{14}$	$1+u^{29}$	$1+u^{32}$
$[[680,20,(\leq 15,\leq 15)]]$	34	0.029	$1+u^9$	$1+u^{20}$	$1+u^{22}$	$1+u^{23}$	$1+u^{26}$	$1+u^{27}$
$[[700,20,(\leq 15,\leq 15)]]$	35	0.029	$1+u^2$	$1+u^{10}$	$1+u^{16}$	$1+u^{23}$	$1+u^{27}$	$1+u^{32}$
$[[720,20,(\leq 15,\leq 15)]]$	36	0.028	$1+u^4$	$1+u^{14}$	$1+u^{15}$	$1+u^{19}$	$1+u^{25}$	$1+u^{31}$
$[[740,20,(\leq 18,\leq 18)]]$	37	0.027	$1+u^5$	$1+u^{11}$	$1+u^{20}$	$1+u^{23}$	$1+u^{33}$	$1+u^{35}$
$[[760,20,(\leq 18,\leq 18)]]$	38	0.026	$1+u^{22}$	$1+u^{24}$	$1+u^{25}$	$1+u^{33}$	$1+u^{34}$	$1+u^{37}$
$[[780,20,(\leq 18,\leq 18)]]$	39	0.026	$1+u^{14}$	$1+u^{15}$	$1+u^{19}$	$1+u^{23}$	$1+u^{26}$	$1+u^{30}$
$[[800,20,(\leq 15,\leq 15)]]$	40	0.025	$1+u^3$	$1+u^{13}$	$1+u^{14}$	$1+u^{15}$	$1+u^{22}$	$1+u^{30}$
$[[820,20,(\leq 18,\leq 18)]]$	41	0.024	$1+u^2$	$1+u^3$	$1+u^4$	$1+u^9$	$1+u^{16}$	$1+u^{26}$
$[[840,20,(\leq 18,\leq 18)]]$	42	0.024	$1+u^{13}$	$1+u^{15}$	$1+u^{18}$	$1+u^{19}$	$1+u^{35}$	$1+u^{40}$
$[[860,20,(\leq 20,\leq 20)]]$	43	0.023	$1+u^3$	$1+u^{14}$	$1+u^{16}$	$1+u^{17}$	$1+u^{25}$	$1+u^{38}$
$[[880,20,(\leq 15,\leq 15)]]$	44	0.023	$1+u^{13}$	$1+u^{20}$	$1+u^{21}$	$1+u^{25}$	$1+u^{26}$	$1+u^{41}$
$[[900,20,(\leq 18,\leq 18)]]$	45	0.022	$1+u^7$	$1+u^{17}$	$1+u^{21}$	$1+u^{22}$	$1+u^{32}$	$1+u^{43}$
$[[920,20,(\leq 18,\leq 18)]]$	46	0.022	$1+u^3$	$1+u^{31}$	$1+u^{33}$	$1+u^{34}$	$1+u^{36}$	$1+u^{45}$
$[[940,20,(\leq 18,\leq 18)]]$	47	0.021	$1+u^{11}$	$1+u^{14}$	$1+u^{23}$	$1+u^{32}$	$1+u^{37}$	$1+u^{44}$
$[[960,20,(\leq 19,\leq 19)]]$	48	0.021	$1+u^7$	$1+u^{10}$	$1+u^{11}$	$1+u^{33}$	$1+u^{34}$	$1+u^{47}$
$[[980,20,(\leq 19,\leq 19)]]$	49	0.020	$1+u^6$	$1+u^9$	$1+u^{11}$	$1+u^{14}$	$1+u^{32}$	$1+u^{48}$

TABLE VIII (continued)

$[[n,k,(\leq d_X,\leq d_Z)]]$	ℓ	k/n	F	G	H	I	J	K
$[[1000,20,(\leq 20,\leq 20)]]$	50	0.020	$1+u^{13}$	$1+u^{17}$	$1+u^{19}$	$1+u^{23}$	$1+u^{39}$	$1+u^{41}$
$[[1020,20,(\leq 19,\leq 19)]]$	51	0.020	$1+u^{16}$	$1+u^{26}$	$1+u^{27}$	$1+u^{30}$	$1+u^{31}$	$1+u^{44}$
$[[1040,20,(\leq 18,\leq 18)]]$	52	0.019	$1+u^{4}$	$1+u^{21}$	$1+u^{35}$	$1+u^{42}$	$1+u^{47}$	$1+u^{50}$
$[[1060,20,(\leq 20,\leq 20)]]$	53	0.019	$1+u^{6}$	$1+u^{25}$	$1+u^{32}$	$1+u^{43}$	$1+u^{44}$	$1+u^{49}$
$[[1080,20,(\leq 20,\leq 20)]]$	54	0.019	$1+u^{5}$	$1+u^{32}$	$1+u^{34}$	$1+u^{38}$	$1+u^{41}$	$1+u^{53}$
$[[1100,20,(\leq 21,\leq 21)]]$	55	0.018	$1+u^{6}$	$1+u^{25}$	$1+u^{27}$	$1+u^{31}$	$1+u^{32}$	$1+u^{35}$
$[[1120,20,(\leq 20,\leq 20)]]$	56	0.018	$1+u$	$1+u^{10}$	$1+u^{14}$	$1+u^{26}$	$1+u^{34}$	$1+u^{37}$
$[[1140,20,(\leq 21,\leq 21)]]$	57	0.018	$1+u^{4}$	$1+u^{11}$	$1+u^{25}$	$1+u^{29}$	$1+u^{44}$	$1+u^{48}$
$[[1160,20,(\leq 21,\leq 21)]]$	58	0.017	$1+u^{4}$	$1+u^{11}$	$1+u^{17}$	$1+u^{25}$	$1+u^{48}$	$1+u^{52}$
$[[1180,20,(\leq 22,\leq 22)]]$	59	0.017	$1+u^{14}$	$1+u^{29}$	$1+u^{36}$	$1+u^{40}$	$1+u^{47}$	$1+u^{51}$
$[[1200,20,(\leq 20,\leq 20)]]$	60	0.017	$1+u^{3}$	$1+u^{18}$	$1+u^{27}$	$1+u^{52}$	$1+u^{53}$	$1+u^{55}$
$[[1220,20,(\leq 20,\leq 20)]]$	61	0.016	$1+u^{10}$	$1+u^{19}$	$1+u^{26}$	$1+u^{29}$	$1+u^{46}$	$1+u^{49}$
$[[1240,20,(\leq 24,\leq 24)]]$	62	0.016	$1+u^{4}$	$1+u^{17}$	$1+u^{35}$	$1+u^{51}$	$1+u^{53}$	$1+u^{59}$
$[[1260,20,(\leq 21,\leq 21)]]$	63	0.016	$1+u$	$1+u^{7}$	$1+u^{20}$	$1+u^{37}$	$1+u^{51}$	$1+u^{60}$
$[[1280,20,(\leq 23,\leq 23)]]$	64	0.016	$1+u^{15}$	$1+u^{20}$	$1+u^{29}$	$1+u^{38}$	$1+u^{48}$	$1+u^{57}$
$[[1300,20,(\leq 23,\leq 23)]]$	65	0.015	$1+u^{12}$	$1+u^{20}$	$1+u^{22}$	$1+u^{24}$	$1+u^{26}$	$1+u^{35}$
$[[1320,20,(\leq 22,\leq 22)]]$	66	0.015	$1+u^{18}$	$1+u^{25}$	$1+u^{31}$	$1+u^{39}$	$1+u^{45}$	$1+u^{47}$

TABLE VIII: **Collapsed 6-dimensional (6D) multivariate multicycle (MM) codes.** The multivariate quotient ring polynomials $F(u,v,w,x,y,z)$, $G(u,v,w,x,y,z)$, $H(u,v,w,x,y,z)$, $I(u,v,w,x,y,z)$, $J(u,v,w,x,y,z)$ and $K(u,v,w,x,y,z)$ are defined over $\mathbb{F}_2[u,v,w,x,y,z]/\langle u^\ell-1, v-1, w-1, x-1, y-1, z-1\rangle$. All these 6D MM codes have weight-8 X stabilizer checks and weight-8 Z stabilizer checks.

$[[n,k,(\leq d_X,\leq d_Z)]]$	ℓ	k/n	F	G	H	I	J	K	L
$[[385,35,(\leq 10,\leq 7)]]$	11	0.091	$1+\tau$	$1+\tau^{2}$	$1+\tau^{3}$	$1+\tau^{4}$	$1+\tau^{6}$	$1+\tau^{8}$	$1+\tau^{10}$
$[[455,35,(\leq 10,\leq 7)]]$	13	0.077	$1+\tau^{2}$	$1+\tau^{6}$	$1+\tau^{7}$	$1+\tau^{8}$	$1+\tau^{9}$	$1+\tau^{10}$	$1+\tau^{11}$
$[[455,35,(\leq 11,\leq 7)]]$	13	0.077	$1+\tau$	$1+\tau^{2}$	$1+\tau^{4}$	$1+\tau^{5}$	$1+\tau^{6}$	$1+\tau^{10}$	$1+\tau^{12}$
$[[490,35,(\leq 11,\leq 7)]]$	14	0.071	$1+\tau$	$1+\tau^{3}$	$1+\tau^{7}$	$1+\tau^{8}$	$1+\tau^{9}$	$1+\tau^{11}$	$1+\tau^{12}$
$[[490,35,(\leq 12,\leq 7)]]$	14	0.071	$1+\tau^{3}$	$1+\tau^{4}$	$1+\tau^{6}$	$1+\tau^{7}$	$1+\tau^{9}$	$1+\tau^{12}$	$1+\tau^{13}$
$[[525,35,(\leq 12,\leq 8)]]$	15	0.067	$1+\tau^{2}$	$1+\tau^{3}$	$1+\tau^{6}$	$1+\tau^{7}$	$1+\tau^{8}$	$1+\tau^{10}$	$1+\tau^{11}$
$[[525,35,(\leq 15,\leq 9)]]$	15	0.067	$1+\tau$	$1+\tau^{2}$	$1+\tau^{3}$	$1+\tau^{4}$	$1+\tau^{5}$	$1+\tau^{8}$	$1+\tau^{9}$
$[[560,35,(\leq 15,\leq 8)]]$	16	0.062	$1+\tau^{2}$	$1+\tau^{4}$	$1+\tau^{6}$	$1+\tau^{7}$	$1+\tau^{11}$	$1+\tau^{13}$	$1+\tau^{15}$
$[[595,35,(\leq 15,\leq 10)]]$	17	0.059	$1+\tau^{2}$	$1+\tau^{5}$	$1+\tau^{8}$	$1+\tau^{10}$	$1+\tau^{11}$	$1+\tau^{13}$	$1+\tau^{16}$
$[[630,35,(\leq 15,\leq 9)]]$	18	0.056	$1+\tau^{3}$	$1+\tau^{4}$	$1+\tau^{6}$	$1+\tau^{8}$	$1+\tau^{11}$	$1+\tau^{13}$	$1+\tau^{17}$
$[[665,35,(\leq 15,\leq 10)]]$	19	0.053	$1+\tau^{8}$	$1+\tau^{9}$	$1+\tau^{13}$	$1+\tau^{14}$	$1+\tau^{16}$	$1+\tau^{17}$	$1+\tau^{18}$
$[[700,35,(\leq 16,\leq 10)]]$	20	0.050	$1+\tau^{6}$	$1+\tau^{9}$	$1+\tau^{12}$	$1+\tau^{13}$	$1+\tau^{17}$	$1+\tau^{18}$	$1+\tau^{19}$
$[[735,35,(\leq 15,\leq 10)]]$	21	0.048	$1+\tau$	$1+\tau^{5}$	$1+\tau^{6}$	$1+\tau^{11}$	$1+\tau^{13}$	$1+\tau^{17}$	$1+\tau^{18}$
$[[770,35,(\leq 17,\leq 10)]]$	22	0.045	$1+\tau$	$1+\tau^{5}$	$1+\tau^{7}$	$1+\tau^{12}$	$1+\tau^{14}$	$1+\tau^{16}$	$1+\tau^{19}$
$[[805,35,(\leq 18,\leq 11)]]$	23	0.043	$1+\tau^{4}$	$1+\tau^{7}$	$1+\tau^{10}$	$1+\tau^{12}$	$1+\tau^{18}$	$1+\tau^{20}$	$1+\tau^{22}$
$[[840,35,(\leq 19,\leq 12)]]$	24	0.042	$1+\tau^{2}$	$1+\tau^{5}$	$1+\tau^{7}$	$1+\tau^{13}$	$1+\tau^{14}$	$1+\tau^{20}$	$1+\tau^{23}$

TABLE IX (continued)

$[[n,k,(\leq d_X,\leq d_Z)]]$	ℓ	k/n	F	G	H	I	J	K	L
$[[875,35,(\leq 18,\leq 11)]]$	25	0.040	$1+\tau^2$	$1+\tau^3$	$1+\tau^8$	$1+\tau^{11}$	$1+\tau^{13}$	$1+\tau^{18}$	$1+\tau^{20}$
$[[910,35,(\leq 19,\leq 11)]]$	26	0.038	$1+\tau^2$	$1+\tau^6$	$1+\tau^9$	$1+\tau^{15}$	$1+\tau^{16}$	$1+\tau^{23}$	$1+\tau^{25}$
$[[945,35,(\leq 18,\leq 9)]]$	27	0.037	$1+\tau^6$	$1+\tau^7$	$1+\tau^9$	$1+\tau^{10}$	$1+\tau^{12}$	$1+\tau^{16}$	$1+\tau^{23}$
$[[980,35,(\leq 20,\leq 12)]]$	28	0.036	$1+\tau$	$1+\tau^4$	$1+\tau^{10}$	$1+\tau^{13}$	$1+\tau^{17}$	$1+\tau^{18}$	$1+\tau^{25}$
$[[1015,35,(\leq 22,\leq 13)]]$	29	0.034	$1+\tau^8$	$1+\tau^{13}$	$1+\tau^{18}$	$1+\tau^{19}$	$1+\tau^{23}$	$1+\tau^{25}$	$1+\tau^{28}$
$[[1050,35,(\leq 20,\leq 13)]]$	30	0.033	$1+\tau^2$	$1+\tau^5$	$1+\tau^7$	$1+\tau^{13}$	$1+\tau^{21}$	$1+\tau^{24}$	$1+\tau^{29}$

TABLE IX: **Collapsed 7-dimensional (7D) multivariate multicycle (MM) codes.** The multivariate quotient ring polynomials $F(\tau,u,v,w,x,y,z)$, $G(\tau,u,v,w,x,y,z)$, $H(\tau,u,v,w,x,y,z)$, $I(\tau,u,v,w,x,y,z)$, $J(\tau,u,v,w,x,y,z)$, $K(\tau,u,v,w,x,y,z)$ and $L(\tau,u,v,w,x,y,z)$ are defined over $\mathbb{F}_2[\tau,u,v,w,x,y,z]/\langle\tau^\ell-1,u-1,v-1,w-1,x-1,y-1,z-1\rangle$. All these 7D MM codes have weight-10 X stabilizer checks and weight-8 Z stabilizer checks.

$[[n,k,(\leq d_X,\leq d_Z)]]$	ℓ	k/n	F	G	H	I	J	K	L	M
$[[630,70,(9,9)]]$	9	0.111	$1+s$	$1+s^2$	$1+s^3$	$1+s^4$	$1+s^5$	$1+s^6$	$1+s^7$	$1+s^8$
$[[700,70,(\leq 8,\leq 8)]]$	10	0.100	$1+s$	$1+s^2$	$1+s^3$	$1+s^4$	$1+s^5$	$1+s^6$	$1+s^7$	$1+s^9$
$[[770,70,(\leq 9,\leq 9)]]$	11	0.091	$1+s$	$1+s^2$	$1+s^4$	$1+s^6$	$1+s^7$	$1+s^8$	$1+s^9$	$1+s^{10}$
$[[840,70,(\leq 10,\leq 10)]]$	12	0.083	$1+s$	$1+s^2$	$1+s^3$	$1+s^4$	$1+s^5$	$1+s^7$	$1+s^9$	$1+s^{11}$
$[[910,70,(\leq 11,\leq 11)]]$	13	0.077	$1+s^2$	$1+s^3$	$1+s^6$	$1+s^8$	$1+s^9$	$1+s^{10}$	$1+s^{11}$	$1+s^{12}$
$[[980,70,(\leq 10,\leq 10)]]$	14	0.071	$1+s^2$	$1+s^3$	$1+s^7$	$1+s^9$	$1+s^{10}$	$1+s^{11}$	$1+s^{12}$	$1+s^{13}$
$[[1050,70,(\leq 11,\leq 11)]]$	15	0.067	$1+s$	$1+s^2$	$1+s^4$	$1+s^5$	$1+s^6$	$1+s^8$	$1+s^{12}$	$1+s^{14}$
$[[1120,70,(\leq 11,\leq 11)]]$	16	0.062	$1+s$	$1+s^2$	$1+s^3$	$1+s^4$	$1+s^7$	$1+s^9$	$1+s^{11}$	$1+s^{15}$
$[[1190,70,(\leq 13,\leq 13)]]$	17	0.059	$1+s$	$1+s^3$	$1+s^4$	$1+s^5$	$1+s^9$	$1+s^{10}$	$1+s^{12}$	$1+s^{16}$
$[[1260,70,(\leq 12,\leq 12)]]$	18	0.056	$1+s^4$	$1+s^5$	$1+s^6$	$1+s^7$	$1+s^{11}$	$1+s^{13}$	$1+s^{15}$	$1+s^{16}$
$[[1330,70,(\leq 13,\leq 13)]]$	19	0.053	$1+s^3$	$1+s^6$	$1+s^8$	$1+s^{10}$	$1+s^{13}$	$1+s^{14}$	$1+s^{15}$	$1+s^{17}$
$[[1400,70,(\leq 10,\leq 10)]]$	20	0.050	$1+s^3$	$1+s^4$	$1+s^7$	$1+s^9$	$1+s^{10}$	$1+s^{11}$	$1+s^{14}$	$1+s^{17}$
$[[1470,70,(\leq 13,\leq 13)]]$	21	0.048	$1+s^2$	$1+s^4$	$1+s^5$	$1+s^9$	$1+s^{11}$	$1+s^{15}$	$1+s^{17}$	$1+s^{20}$
$[[1540,70,(\leq 12,\leq 12)]]$	22	0.045	$1+s^3$	$1+s^5$	$1+s^7$	$1+s^8$	$1+s^{14}$	$1+s^{15}$	$1+s^{17}$	$1+s^{20}$
$[[1610,70,(\leq 16,\leq 16)]]$	23	0.043	$1+s$	$1+s^2$	$1+s^3$	$1+s^6$	$1+s^7$	$1+s^8$	$1+s^{12}$	$1+s^{13}$
$[[1680,70,(\leq 15,\leq 15)]]$	24	0.042	$1+s$	$1+s^6$	$1+s^9$	$1+s^{13}$	$1+s^{15}$	$1+s^{16}$	$1+s^{19}$	$1+s^{22}$
$[[1750,70,(\leq 17,\leq 17)]]$	25	0.040	$1+s^3$	$1+s^5$	$1+s^7$	$1+s^{16}$	$1+s^{17}$	$1+s^{19}$	$1+s^{21}$	$1+s^{23}$
$[[1820,70,(\leq 13,\leq 13)]]$	26	0.038	$1+s^3$	$1+s^9$	$1+s^{10}$	$1+s^{12}$	$1+s^{15}$	$1+s^{18}$	$1+s^{19}$	$1+s^{22}$
$[[1890,70,(\leq 15,\leq 15)]]$	27	0.037	$1+s$	$1+s^5$	$1+s^8$	$1+s^{11}$	$1+s^{13}$	$1+s^{18}$	$1+s^{23}$	$1+s^{24}$
$[[1960,70,(\leq 14,\leq 14)]]$	28	0.036	$1+s^3$	$1+s^{11}$	$1+s^{12}$	$1+s^{17}$	$1+s^{18}$	$1+s^{20}$	$1+s^{23}$	$1+s^{26}$
$[[2030,70,(\leq 18,\leq 18)]]$	29	0.034	$1+s^4$	$1+s^7$	$1+s^{11}$	$1+s^{13}$	$1+s^{14}$	$1+s^{17}$	$1+s^{20}$	$1+s^{28}$

TABLE X: **Collapsed 8-dimensional (8D) multivariate multicycle (MM) codes.** The multivariate quotient ring polynomials $F(s,\tau,u,v,w,x,y,z)$, $G(s,\tau,u,v,w,x,y,z)$, $H(s,\tau,u,v,w,x,y,z)$, $I(s,\tau,u,v,w,x,y,z)$, $J(s,\tau,u,v,w,x,y,z)$, $K(s,\tau,u,v,w,x,y,z)$, $L(s,\tau,u,v,w,x,y,z)$ and $M(s,\tau,u,v,w,x,y,z)$ are defined over $\mathbb{F}_2[s,\tau,u,v,w,x,y,z]/\langle s^\ell-1,\tau-1,u-1,v-1,w-1,x-1,y-1,z-1\rangle$. All these 8D MM codes have weight-10 X stabilizer checks and weight-10 Z stabilizer checks.

$[[n,k,(\leq d_X,\leq d_Z)]]$	ℓ	k/n	F	G	H	I	J	K	L	M	N
$[[1386,126,(\leq 11,\leq 9)]]$	11	0.091	$1+r^2$	$1+r^3$	$1+r^4$	$1+r^5$	$1+r^6$	$1+r^7$	$1+r^8$	$1+r^9$	$1+r^{10}$
$[[1512,126,(\leq 12,\leq 9)]]$	12	0.083	$1+r$	$1+r^2$	$1+r^3$	$1+r^5$	$1+r^7$	$1+r^8$	$1+r^9$	$1+r^{10}$	$1+r^{11}$
$[[1638,126,(\leq 13,\leq 10)]]$	13	0.077	$1+r$	$1+r^2$	$1+r^4$	$1+r^5$	$1+r^6$	$1+r^7$	$1+r^8$	$1+r^{10}$	$1+r^{11}$
$[[1764,126,(\leq 14,\leq 10)]]$	14	0.071	$1+r$	$1+r^2$	$1+r^3$	$1+r^4$	$1+r^5$	$1+r^6$	$1+r^7$	$1+r^9$	$1+r^{13}$

TABLE XI: **Collapsed 9-dimensional (9D) multivariate multicycle (MM) codes.** All these 9D MM codes have weight-12 X stabilizer checks and weight-10 Z stabilizer checks. The multivariate quotient ring polynomials $F(r,s,\tau,u,v,w,x,y,z)$, $G(r,s,\tau,u,v,w,x,y,z)$, $H(r,s,\tau,u,v,w,x,y,z)$, $I(r,s,\tau,u,v,w,x,y,z)$, $J(r,s,\tau,u,v,w,x,y,z)$, $K(r,s,\tau,u,v,w,x,y,z)$, $L(r,s,\tau,u,v,w,x,y,z)$, $M(r,s,\tau,u,v,w,x,y,z)$ and $N(r,s,\tau,u,v,w,x,y,z)$ are defined over $\mathbb{F}_2[r,s,\tau,u,v,w,x,y,z]/\langle r^\ell-1, s-1,\tau-1,u-1,v-1,w-1,x-1,y-1,z-1\rangle$.